

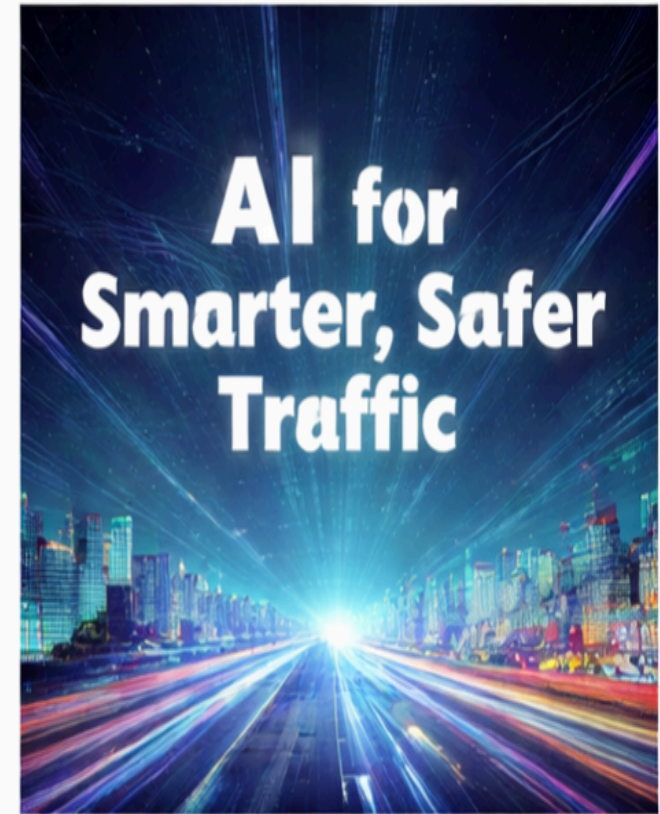
Nameless: smarter, safer Traffic

Venture Viability Analysis
Sri Sairam Engineering College

Nameless: smarter, safer Traffic

We build AI systems that clear traffic and create fast routes for emergency vehicles.

Sri Sairam Engineering College



A photograph of three men in an office setting. They are gathered around a desk, looking at a laptop. The man on the left is wearing a blue shirt and is typing on the laptop. The man in the middle is wearing a grey shirt and glasses, smiling at the camera. The man on the right is wearing a white shirt and is also smiling. In the background, there are whiteboards with diagrams and charts.

Context

Bengaluru drivers lose 117 hours/year in traffic; ambulance delays in Chennai rise by 75%.

Problem Statement

Problem

Urban traffic systems rely on fixed-time signals that cannot adapt to real-time traffic, causing congestion, long delays, fuel waste, and slow emergen

Impact

Commuters face long delays and fuel costs, public transport runs late, emergency vehicles are slowed, and city authorities struggle with congestion, p



Problem Statement/Industry

Problem Being Solved

Urban traffic systems are inefficient because most cities rely on fixed-time traffic signals that cannot adapt to real-time traffic conditions. This leads to severe congestion, longer commute times, wasted fuel, delays in public transport, and slower emergency response, creating stress for commuters and safety risks in rapidly growing cities.

Supporting Data

Bengaluru drivers lose 117 hours/year in traffic; ambulance delays in Chennai rise by 75%.

Source: TomTom Traffic Index 2025; Times of India Traffic & Ambulance Speed Analysis Report.



Area
Mobility



Industry
Transportation & Logistics



Domain
Intelligent Transportation
Systems

Problem Analysis



Affected Stakeholders

Urban commuters, public transport operators, emergency services (ambulances and fire departments), city traffic management authorities, and urban municipal governments are affected by inefficient traffic systems. Traffic congestion causes delays, higher fuel costs, increased stress for commuters, disrupted transport schedules, and slower emergency



Impact on Stakeholders

Commuters experience longer travel times, higher fuel expenses, and daily stress. Public transport operators face delays and reduced schedule reliability. Emergency services struggle to reach destinations quickly, risking lives. City authorities deal with increased congestion, pollution, economic losses, and public dissatisfaction with traffic management



Root Causes

Outdated fixed-time traffic signals, lack of real-time traffic monitoring, poor coordination between intersections, and limited use of data-driven traffic management contribute to the problem. Rapid urbanization, increasing vehicle numbers, and inadequate road infrastructure further worsen congestion and delays in cities.



Personal/Team Connect

As engineering students, we frequently experience heavy traffic and delays in our daily commute. We have seen how congestion wastes time, increases stress, and delays emergency vehicles. With our skills in technology and programming, we are motivated to work on this problem and contribute to improving urban mobility and public safety.

Target Customer Segments

Primary

Urban municipal traffic management authorities in major metropolitan cities.



Secondary

Smart Traffic Management System

Customer Segment & Persona

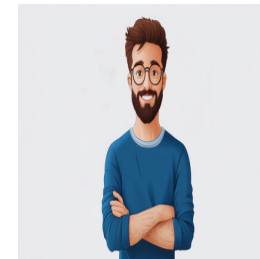
Primary Segment

Urban municipal traffic management authorities in major metropolitan cities.

Secondary Segment

Smart Traffic Management System

Persona



Hendry

Age in years: 36

Location: Metropolitan

Organizational Role: Traffic operations and urban mobility planning manager.
(if applicable)

Customer Profile

Education: College pass out

Gender: Male

Occupation: Works for a private company

Interests/Hobbies: Travel

Primary Source of Information: News Apps

Shopping Preference: Mostly Online

Comfort with Technology: High

Favourite Social Media: Instagram

Favourite Offline Gathering Spots: Private Workspace

Jobs-to-be-Done

Functional JTBD



Reduce traffic congestion, manage traffic signals efficiently, monitor real-time traffic conditions, improve vehicle flow at intersections, ensure faster emergency vehicle movement, and maintain reliable public transport schedules across the city.

Emotional JTBD



Feel confident in managing city traffic effectively, reduce stress from public complaints about congestion, feel in control of traffic situations, and gain satisfaction from ensuring safer roads and faster emergency response across the city.

Social JTBD



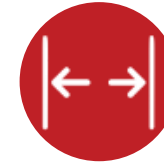
Gain recognition from city leaders and the public for improving traffic conditions, earn trust from citizens for managing roads efficiently, build a reputation as a forward-thinking urban mobility leader, receive appreciation for reducing congestion and enabling faster emergency response in the city's transportation system.

Current Alternatives



Current Alternatives

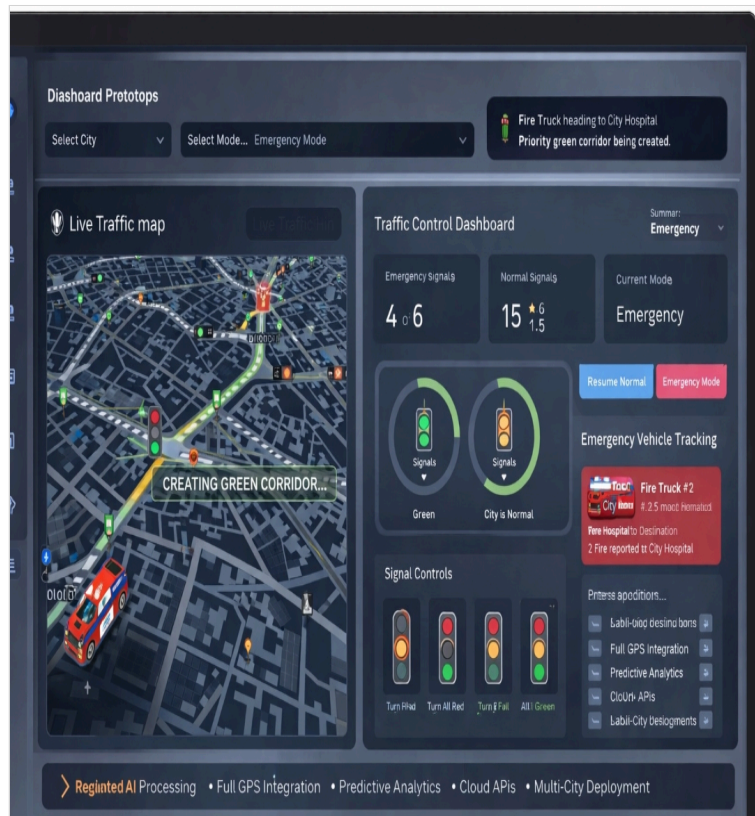
Traffic authorities currently manage congestion using fixed-time traffic signals, manual traffic control by police during peak hours, CCTV monitoring systems, and occasional traffic surveys. They also rely on basic sensor data and manual coordination between intersections to regulate vehicle flow and handle traffic during emergencies or heavy conge



Gaps in Current Alternatives

Existing solutions rely on fixed signal timings and manual control, which cannot adapt to changing traffic conditions. CCTV and basic sensors provide limited data, making real-time decisions difficult. Manual monitoring is time-consuming and inefficient, leading to congestion, delayed emergency vehicles, and dissatisfaction among citizens and city

Our Solution



Solution

3. Emergency Vehicle Priority Network Connects ambulances/fire trucks to signals via GPS to automatically clear their route.

Core Technologies/ Methodologies

Python, TensorFlow AI, GPS tracking, IoT sensors, React.js, Node.js, PostgreSQL, cloud APIs for signal integration

Solution Design



Our Solution

3. Emergency Vehicle Priority Network
Connects ambulances/fire trucks to signals via GPS to automatically clear their route.



Key Features

Real-time signal control, emergency green corridors, GPS-based vehicle tracking, predictive congestion alerts, central dashboard for traffic monitoring



Uniqueness

AI-driven real-time adaptive signals with automatic emergency routing, optimized for Indian traffic conditions, scalable SaaS model, faster deployment

Solution Format:

Digital Service (AI Traffic Management SaaS)

Core Technologies/ Methodologies:

Python, TensorFlow AI, GPS tracking, IoT sensors, React.js, Node.js, PostgreSQL, cloud APIs for signal integration

Solution Benefits



Functional Benefits

Faster emergency response, reduced congestion, optimized signal timing, lower fuel use, improved traffic flow, better data insights, and efficient traffic management.



Emotional Benefits

Peace of mind from quick emergency response, less stress from traffic, confidence in smart systems, and satisfaction in improved city safety.



Social Benefits

Public appreciation, recognition as a smart city, trust in authorities, improved city image, and increased civic pride among citizens.



Macro Benefits

Reduces emissions and fuel waste, improves safety, saves time, boosts productivity, and supports sustainable smart city development.

Competitors



Direct

Siemens Mobility, SWARCO, Cubic Transportation Systems, Kapsch TrafficCom
Siemens Mobility, SWARCO, Kapsch TrafficCom, Cubic, Econolite



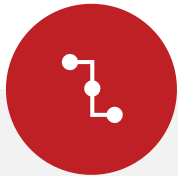
Indirect

Google Maps, Waze, Uber, Ola, Indian traffic management systems
Google, Uber, Waze, Citymapper, Moovit

Our UVP

AI traffic signals with real-time control and instant emergency green corridors to save lives

Competitors



Direct **Competitors**

Siemens Mobility,
SWARCO, Cubic
Transportation Systems,
Kapsch TrafficCom



Indirect **Competitors**

Google Maps, Waze,
Uber, Ola, Indian traffic
management systems



Direct **Competitors** **Globally**

Siemens Mobility,
SWARCO, Kapsch
TrafficCom, Cubic,
Econolite



Indirect **Competitors** **Globally**

Google, Uber, Waze,
Citymapper, Moovit

Macro Analysis

Favourable Trends

AREA	DESCRIPTION
Other	1. Growth of Smart Cities Rising smart city projects boost demand for AI-based traffic solutions.
Other	2. IoT and 5G Expansion Faster connectivity enables real-time traffic data and smart signal control.

Unfavourable Trends

AREA	DESCRIPTION
Other	1. Government Procurement Delays Slow approvals and complex tenders delay project implementation.
Other	2. Budget Constraints Limited city budgets restrict spending on new traffic technologies.

Data Sources:

EY Infrastructure Report 2023, Ministry of Urban Development Reports, Statista Analysis

Back-of-the-Envelope Financial Projections



Currency: Indian Rupee (INR)

Chosen Business Model: Service Provider

AREA	YEAR 1	YEAR 2	YEAR 3
Revenues	18000000	45000000	100000000
Total Expenses	22000000	35000000	65000000
Profit	-4000000	10000000	35000000

Prototype

Prototype Format

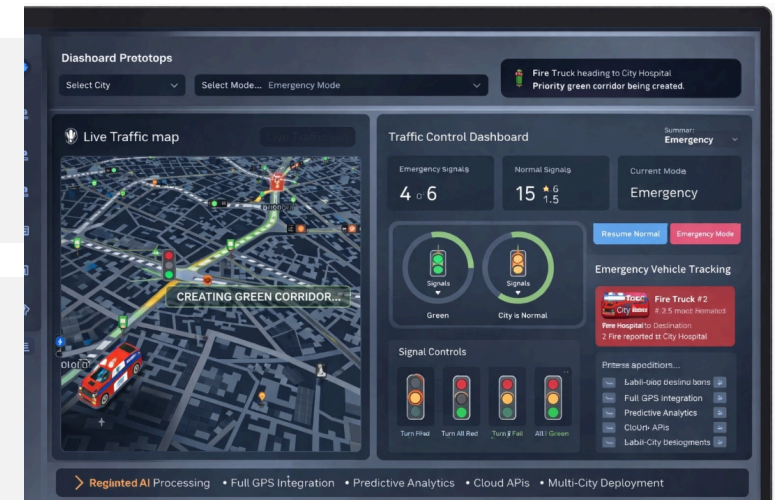
Digital dashboard prototype showing AI traffic control and emergency routing simulation

Functionality included in the Prototype

Live traffic map, signal status, emergency vehicle tracking, and green corridor creation with basic dashboard controls

Functionality NOT included in the Prototype

Real-time AI processing, full GPS integration, predictive analytics, cloud APIs, and large-scale multi-city deployment



{Upload another picture}

Revenue Models / Pricing

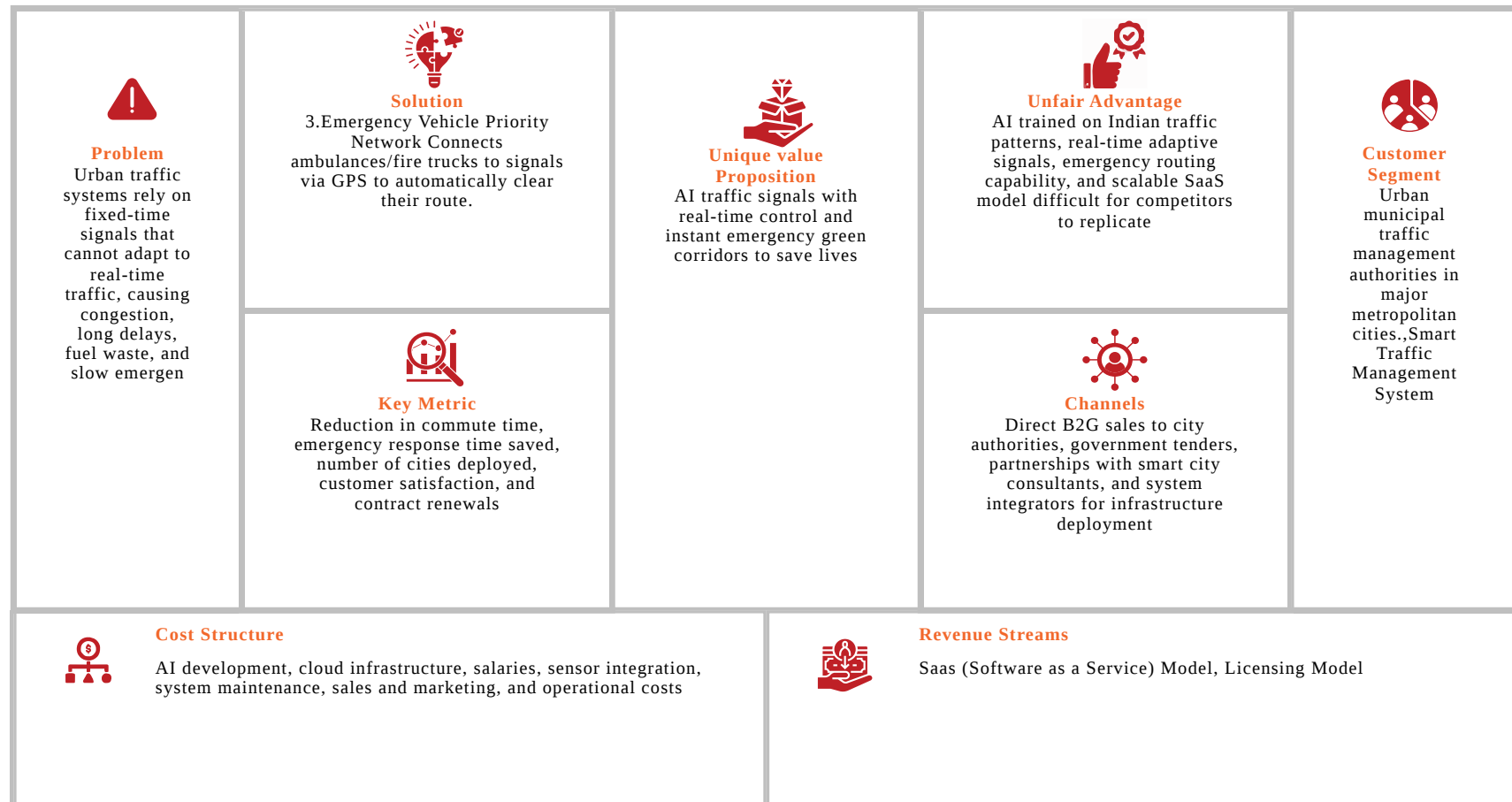
Revenue Model (Primary)

Saas (Software as a Service)
Model

Revenue Model (Secondary)

Licensing Model

Lean Canvas



Financials

Revenue Models/Streams

- SaaS (Software as a Service) Model
- Licensing Model

Pricing

- **Unit of Sale:** Annual SaaS subscription
- **Selling price per unit:**
₹1,00,00,000 per city per year

First Year Projections

Revenues: 13,26,52,929

Operating Profits:
6,46,21,972

Revenue Models / Pricing

Revenue Model (Primary)

SaaS (Software as a Service)
Model

Unit of Sale

Annual SaaS subscription

Sale Price per Unit

₹1,00,00,000 per city per year

Expected units to be sold in Year 1

1 city contract

Expected growth in monthly sales

15% growth per month

Costs & Revenues: Key Assumptions

Currency:

		Year 1 Assumptions	Going from Year 1 to Year 2		Going from Year 2 to Year 3	
TYPE	DESCRIPTION	Financials, Absolute	Business Assumptions	Financial Decisions, % change	Business Assumptions	Financial Decisions, % change
SETUP COSTS	Enter one-time costs like we	5000000	Enter key changes as you go from Year 1 to Year 2, e.g., spend 50% more on advertising Make corresponding changes in financial assumptions in the next column (Example 0: means no change from Year 1, 10 means 10% more from Year 1, -10 means 10% less from Year 1)	0	Enter key changes as you go from Year 2 to Year 3, e.g., hire more people. Make corresponding changes in financial assumptions in the next column (Example 0: means no change from Year 2, 10 means 10% more from Year 2, -10 means 10% less from Year 2)	0
SALES	Enter proposed selling price	18000000		0		10
	Enter the number of units yc	1		200		100
VARIABLE COSTS	Enter total cost of servicing	3000000		50		40
FIXED COSTS (MONTHLY)	Enter salaries of key person	700000		50		40
	Enter the amount you plan t	150000	60	50		
	Enter infrastructure costs th	200000	40	40		
	Enter misc costs, e.g., electr	100000	40	30		

Profit & Loss Projections: Summary

1		Heads	Type	Amount	Plan
2	REVENUES	Service Selling Price			
3		Average Service fee per client	Amount	25000	
4	VARIABLE CO	Service Delivery Costs (per client per month)			
5		Outsourced service support	Amount	3000	
6		Tool Subscriptions (e.g., Canva, Buffer)	Amount	500	
7		Client Support	Amount	1000	
8	FIXED COSTS	Overheads and Salaries (Monthly)			
9		Founder Salaries	Amount	75000	
10		Project Manager salaries	Amount	45000	
11		Other Salaries	Amount	50000	
12		Rent, Administration, Travel, and others	Amount	30000	
13	OTHER FIXED	Other Expenses (Monthly)			
14		Marketing and Sales Team Salaries	Amount	25000	
15		Marketing& Advertising	Amount	20000	
16	FORECAST	Sales Forecast			
17		Client companies in Month #1	Amount	10	
18		Monthly Client Growth Rate% (after chu	Percentage Rate	10	
19		Average client relationship duration in m	Amount	12	
20	INVESTMENT	Investments			
21		Personal Investment (setup costs)	Amount	1000000	
22		Bank loan	Amount	500000	
23		Rate of Interest for the bank loan	Annual Percent	12	
24		Income Tax rate	Annual Percent	20	
25					

Performance & Break-Even Analysis

Year 1 Revenues

13,26,52,929

Gross Profits for Year 1

12,06,52,929

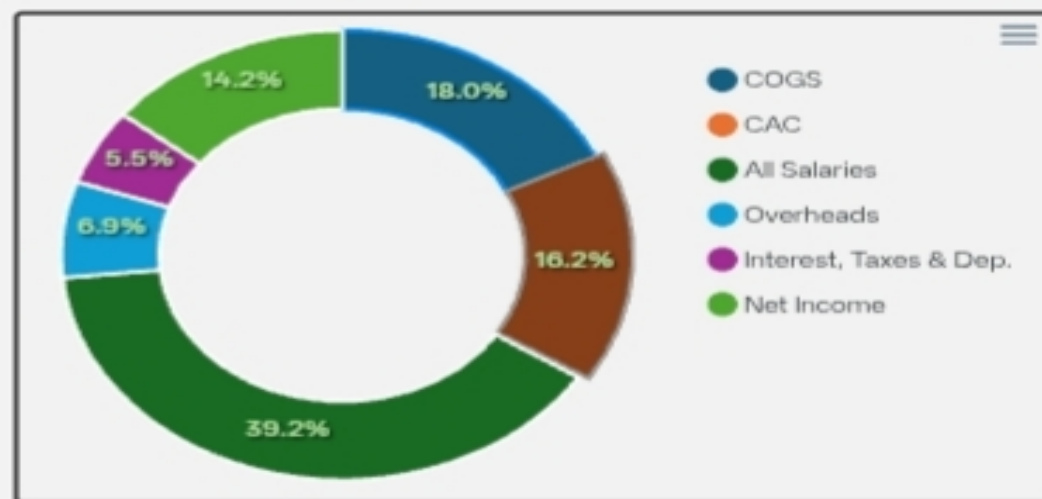
Net Profits for Year 1

6,46,21,972

Break-even Month

Month 1 (August)

Unit Economics





Thank You!

